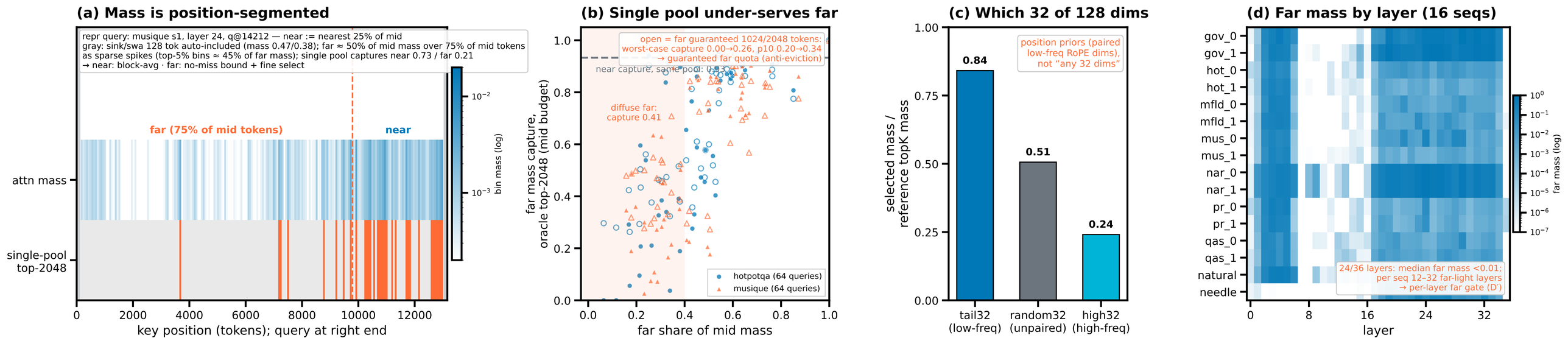

(a) Mass is position-segmented
repr query: musique s1, layer 24, q@14212 — near := nearest 25% of mid
gray: sink/swa 128 tok auto-included (mass 0.47/0.38); far ≈ 50% of mid mass over 75% of mid tokens
as sparse spikes (top-5% bins ≈ 45% of far mass); single pool captures near 0.73 / far 0.21
→ near: block-avg · far: no-miss bound + fine select
far (75% of mid tokens)
near
attn mass
single-pool top-2048
bin mass (log)
key position (tokens); query at right end
(b) Single pool under-serves far
open = far guaranteed 1024/2048 tokens:
worst-case capture 0.00→0.26, p10 0.20→0.34
→ guaranteed far quota (anti-eviction)
near capture, same pool: 0.
diffuse far:
capture 0.41
far mass capture,
oracle top-2048 (mid budget)
far share of mid mass
hotpotqa (64 queries)
musique (64 queries)
(c) Which 32 of 128 dims
position priors (paired
low-freq RoPE dims),
not “any 32 dims”
selected mass /
reference topK mass
0.84
0.51
0.24
tail32
(low-freq)
random32
(unpaired)
high32
(high-freq)
(d) Far mass by layer (16 seqs)
gov_0
gov_1
hot_0
hot_1
mfld_0
mfld_1
mus_0
mus_1
nar_0
nar_1
pr_0
pr_1
qas_0
qas_1
natural
needle
24/36 layers: median far mass <0.01;
per seq 12–32 far-light layers
→ per-layer far gate (D′)
layer
far mass (log)